NOVELTY REPORT — COMPUTATIONAL COUPLING

Status: planning document, superseded. This was written at the start of the project (September 2026), before any experiment was run, and it records the research plan and novelty analysis as they stood then. It is **not** a statement of findings. Several of its premises did not survive: the human H01 microscale tier was withdrawn when its connectivity graph turned out to be empty, the CoupledEventNet / neuromorphic-efficiency direction was withdrawn along with the fabricated silicon figures supporting it, and the central motif hypothesis it sets out was tested and came out with the opposite sign.

For what the project actually found, see `../info.md` and the manuscript. For what was withdrawn and why, see `../AUDIT.md`. This file is kept because the retraction is only checkable against the plan it replaced.

Project: COMPUTATIONAL COUPLING
Lead Autonomous Research Engineer: Antigravity
Audit Date: September 17, 2026
Status: Milestone 0 Initial Audit Complete

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1. Executive Summary & Novelty Classification

We conducted a comprehensive, skeptical audit of the literature across computational neuroscience, network connectomics, machine learning, and neuromorphic engineering to determine whether the proposed research direction is genuinely novel.

Dimension / Candidate Sub-Goal	Classification
A. Cross-species static connectome motif extraction	**RED**
B. Drosophila whole-brain spiking neural simulation	**RED**
C. C. elegans connectome-based AI architectures	**RED**
D. Connectome-informed reservoir computing	**RED**
E. Algorithmic synaptic rewiring in artificial neural nets	**RED**
F. Concept of neural degeneracy	**RED**
G. Human whole-brain macro-scale graph analysis (HCP)	**RED**
H. Connectome-constrained deep networks on sensory tasks	**YELLOW**
I. Event-driven functional abstraction of cross-species conserved motifs	**GREEN**
J. Biologically constrained structural remodeling & the degeneracy frontier (Δ_{struct} vs. F_{task})	**GREEN**
K. Human H01 micro-connectome cross-validation of computational invariants	**GREEN**

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2. Closest Works & Precise Prior Art Analysis

1. Lin, Yang et al. (*Nature* 2024) — "Network statistics of the whole-brain connectome of Drosophila"

- **What it did:** Calculated degree distributions, rich-club curves, clustering coefficients, and 3-node motif counts for FlyWire v630, C. elegans (Cook 2019), zebrafish, and mouse Pinky100.
- **What it left unresolved:** It treated connectomes purely as static topological graphs. It did not simulate, test, or evaluate any computational or cognitive tasks, dynamic signal processing, or event-driven execution.
- **What our work does differently:** We evaluate dynamic *computation*—testing whether these topological structures execute specific temporal and event-driven functions, and whether they can be abstracted into machine learning architectures.

2. Shiu et al. (*Nature* 2024) — "A Drosophila computational brain model reveals sensorimotor processing"

- **What it did:** Implemented a whole-brain LIF model in Python/PyTorch for FlyWire v630 using predicted neurotransmitter signs (Ach, Glu, GABA) to replicate feeding and grooming reflexes.
- **What it left unresolved:** Kept the empirical connectome 100% rigid. Did not ask how the network could be remodeled, did not extract reusable computational modules, and did not evaluate artificial event-stream tasks or energy metrics.
- **What our work does differently:** We use Shiu's LIF model as an upstream verification baseline, but focus on *remodeling* subcircuits and extracting transferable event-driven compute blocks.

3. Shapson-Coe et al. (*Science* 2024) — "A petavoxel fragment of human cerebral cortex reconstructed at nanoscale resolution" (H01)

- **What it did:** Reconstructed a 1 mm^3 volume of human anterior temporal cortex at 4x4x33 nm resolution, identifying ~57k cells, ~150M synapses, axonal whorls, multi-synaptic connections (up to 50 synapses per pair), and laminar E/I distributions.
- **What it left unresolved:** It provided structural connectomics and morphological annotations without functional activity, sensory stimulus co-registration, or computational circuit remodeling.
- **What our work does differently:** We use H01 as a human micro-connectome validation tier to test whether the computational invariants discovered in model organisms persist in human cortical wiring.

4. Suárez, Misic et al. (*Nature Communications* 2024) — "Connectome-based reservoir computing with the conn2res toolbox"

- **What it did:** Evaluated empirical connectomes (C. elegans, mouse, human HCP) as fixed reservoirs for cognitive tasks (XOR, memory capacity, speech classification) using linear readout regression.
- **What it left unresolved:** Connectome topologies were treated as black-box reservoirs with fixed adjacency matrices; no structural remodeling, no evolutionary search, no event-driven asynchronous spike primitives, and no causal motif lesioning.
- **What our work does differently:** We isolate individual motifs, remodel connectivity under biological constraints, and translate them into event-driven spiking architectures rather than relying solely on reservoir readouts.

5. Lechner, Hasani et al. (*Nature Machine Intelligence* 2020, 2021) — "Neural Circuit Policies" (NCPs) & "Liquid Time-Constant Networks" (LTCs)

- **What it did:** Designed a 19-neuron recurrent continuous-time architecture inspired by the 4-layer C. elegans sensorimotor layout for autonomous control.
- **What it left unresolved:** Hand-crafted design; used continuous-time ODE solvers requiring dense integration; did not draw from empirical large-scale connectomes (Drosophila / Mouse / Human) and did not study structural remodeling.

- **What our work does differently:** We ground our architectures in empirical multi-species connectomes (C. elegans, Drosophila, Mouse, Human) and deploy event-driven spiking primitives rather than continuous ODEs.

6. Bellec et al. (*ICLR* 2018) — "Deep Rewiring: Training very sparse neural networks"

- **What it did:** Algorithmically rewired connections during backpropagation in ANNs using random walk / temperature-dependent sampling to maintain extreme parameter sparsity.
- **What it left unresolved:** Completely unconstrained by biology; connections were arbitrary and agnostic to Dale's principle, neurotransmitter signs, or biological motifs.
- **What our work does differently:** We formulate *biologically constrained* remodeling: preserving Dale's principle (E/I sign segregation), cell-type identities, and delay distributions, explicitly tracing the structural distance from biological truth.

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3. Claims We MUST NOT Make (Forbidden Over-Claims)

1. **DO NOT CLAIM:** *"The complete human brain connectome is available."*
   Fact: Only regional macroscale tractography (HCP) and a 1 mm^3 micro-fragment (H01) exist. A complete neuron-level human connectome does not exist.
2. **DO NOT CLAIM:** *"We are the first to compare connectomes across C. elegans, Drosophila, and mouse."*
   Fact: Lin et al. (*Nature* 2024) already published this cross-species comparison.
3. **DO NOT CLAIM:** *"We are the first to simulate the Drosophila FlyWire connectome as a spiking network."*
   Fact: Shiu et al. (*Nature* 2024) already published a whole-brain LIF model.
4. **DO NOT CLAIM:** *"We discovered that biological connectomes have overrepresented feed-forward loops."*
   Fact: Milo et al. (*Science* 2002) and Sporns & Kötter (*PLoS Biol* 2004) proved this over two decades ago.
5. **DO NOT CLAIM:** *"Human brains compute identically to flies or worms."*
   Fact: Scientifically false; different organisms possess unique ecological specializations. We test for conserved computational *primitives*, not universal equivalence.
6. **DO NOT CLAIM:** *"We invented function-preserving rewiring or structural degeneracy."*
   Fact: Marder & Prinz (*Nat Neurosci* 2004) established biological degeneracy; Net2Net (Chen 2016) established functional preservation in deep learning.
7. **DO NOT CLAIM:** *"Brains are fundamentally better than binary computers" or "We created a digital brain/AGI."*
   Fact: Scientifically unjustified and irrelevant to the research questions.

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4. The Defensible, High-Impact Research Question

Primary Research Question:

> **Can neural computational principles identified across biological systems remain meaningful across multiple scales of organization—from invertebrate nervous systems (*C. elegans*) and insect whole brains (*Drosophila*) to mammalian cortical circuits (*Mus musculus*) and human cortical micro-connectivity (H01) and macro-connectivity (HCP)? Specifically: which structural motifs constitute functionally necessary "computational invariants", how far can these circuits be structurally remodeled under biological constraints while preserving dynamic computation, and can their abstractions produce event-driven computing architectures that achieve superior energy-delay efficiency?**

Precise Novel Contributions:

1. **The Biological Degeneracy Frontier Across Scales (Δ_{struct} vs. F_{task}):** First quantitative mapping of how far empirical biological circuits (C. elegans, Drosophila subcircuits, Mouse V1, and human H01) can be structurally deformed under Dale's principle and cell-type constraints before dynamic event-driven computation collapses.
2. **Identification of Computational Invariants vs. Biological Scaffolding:** Causal lesioning and permutation experiments identifying which specific multi-node motifs are computationally non-negotiable (destroying temporal memory or coincidence detection when broken) versus which connections are redundant developmental scaffolding.
3. **Human Micro-Connectome Cross-Validation (H01):** First verification of whether computational invariant motifs identified in model organisms persist in nanoscale human cortical micro-connectivity.
4. **Cross-Scale Event-Driven Architecture (CoupledEventNet / CEN):** A modular, event-driven neural computing architecture composed of abstracted cross-scale motifs, benchmarked against conventional SNNs, LIF reservoirs, and NCPs for energy proxy (synaptic operations), spike sparsity, and temporal classification latency.